UNIVERSITÀ DEGLI STUDI DI UDINE

Laboratorio di Programmazione Web
(laurea in Scienze e Tecnologie Multimediali)

A.A. 2026/2027

Web link alle pagine del corso:

http://elearning.uniud.it/moodle/course/view.php?id=6288

World Wide Web Technologies: overview

Autore:

Carlo Drioli

I materiali in queste slide sono tratti in parte da:

V. Della Mea, L. Di Gaspero, I. Scagnetto,"Programmazione lato server", Apogeo.

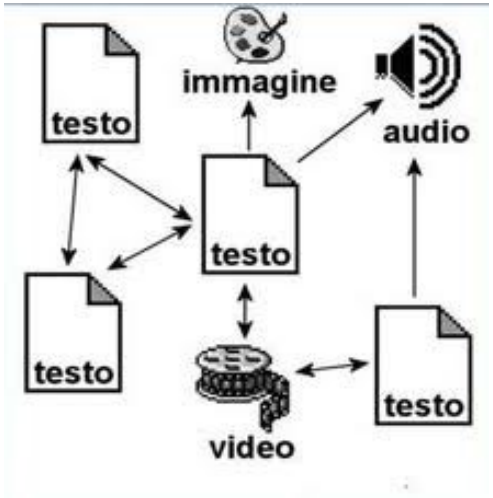
http://www.w3.org/

Wikipedia

The World Wide Web: intro

I. WEB architecture

- The World Wide Web is a *distributed hypermedia* document, consisting of a network of interconnected information resources and hosted by computers located anywhere



- Hypertext: text with references (hyperlinks) to other text that can be immediately accessed

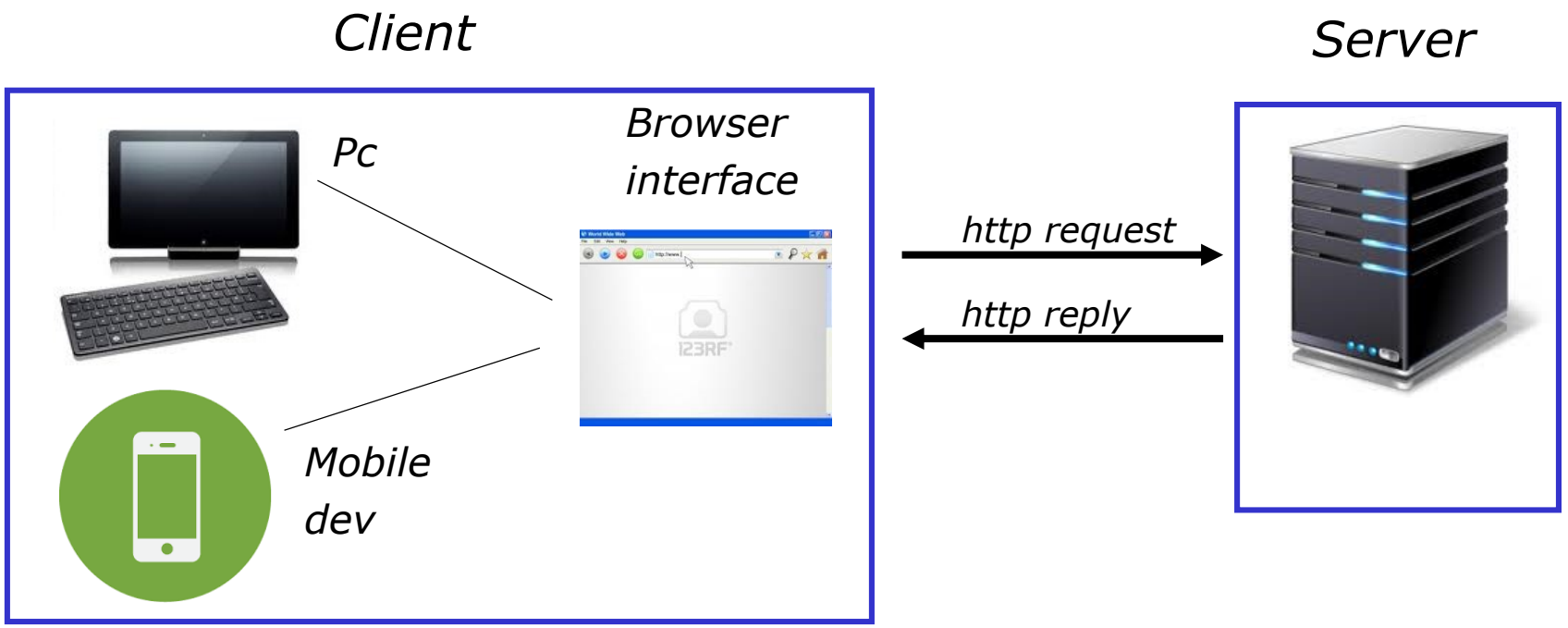
- Multimedia document: a document containing information of heterogeneous nature (text, image, audio, video)

- Hypermedia document: a multimedia document accessible in a non-linear way by using hyperlinks

The World Wide Web: intro

I. WEB architecture

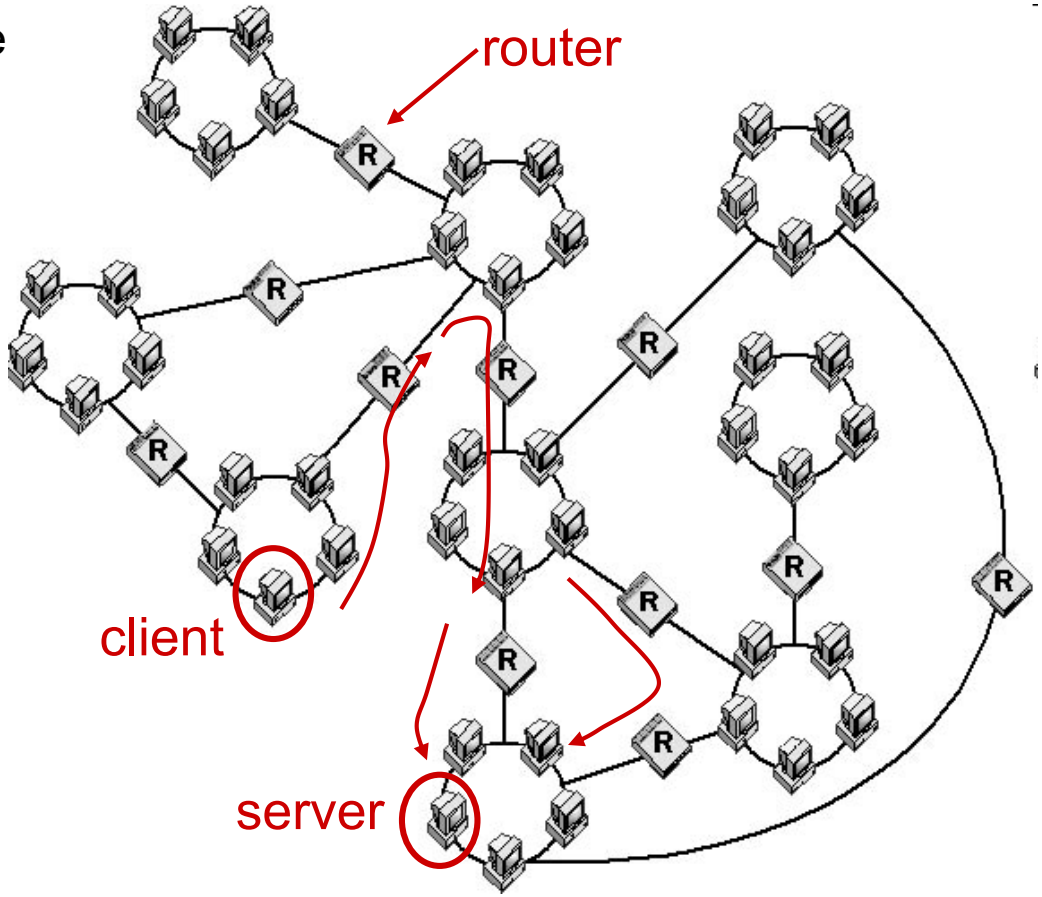
- Access to resources is achieved through *browsers,* a class of applications aimed at communicating with resource hosting systems, called *servers*, via the *client-server* paradigm.



The World Wide Web: intro

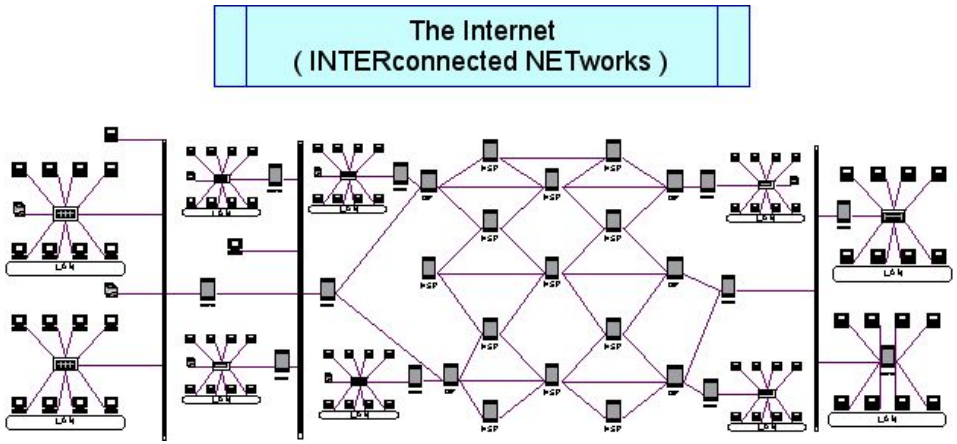
I. WEB architecture

- … where *clients* and *servers* are nodes in a mesh network.



The World Wide Web: intro

I. Network infrastructure: the Internet



- Internet : the interconnection of many heterogeneous networks
- ‘60-’70 : first experiments at DARPA, ARPAnet project implementation
- Packet switching and Ethernet/LAN technologies
- Internet protocols, TCP/IP
- IP: allows sending packets to correct destination through IP addressing
- TCP: organizes data into IP packets, and implement transmission between application softwares

The World Wide Web: intro

II. Web history

- 1989: Tim Berners-Lee, at CERN in Ginevra, wrote a proposal for "a large hypertext database with typed links" (the World Wide Web (WWW) project originally aimed at allowing high energy physicists to share data, news, and documentation)

- 1990: the essential tools are ready. The HyperText Transfer Protocol (HTTP), the HyperText Markup Language (HTML), the first web browser, and HTTP server.

- 1994: the W3C consortium was founded, to create standards and recommendations, promote and manage the growth and development of the www.

The World Wide Web: intro

II. Web history

The W3C:

- develops HTML, CSS, and most Web technology
- Has 380 companies and organizations as members
- Located at MIT (US), Inria (France), Keiko (Japan)
- Organized in working groups
- Manages Web development through
  - Working Drafts, Recommendations, Working Group Notes, etc.

The World Wide Web: intro

II. Web history

- 1994-today:
  - race of competing browsers
  - commercialization of the web and e-commerce
  - search engines
  - ubiquity of the web
  - growth of web applications complexity (web engineering)
  - multimedia data streaming (audio, video) for remote interaction
  - rich multimodal interaction (graphics, audio, speech)
  - social networks
  - semantic web
  - mobile computing (geolocalized web applications)

The World Wide Web: intro

III: Web applications categories and complexity

Characteristics of **simple** vs **advanced** Web applications (1/2)

Simple Web-based systems	Advanced Web-based systems
• Primarily textual information	• Rich multimedia information (audio, music, speech, image, video)
• Information content fairly static	• Dynamic web pages because information changes with user needs
• Simple navigation, limited information and usefulness	• Large volume of information • Difficult to navigate and find information
• Limited interactivity and functionality	• Enhanced multimodal interaction
• Stand alone systems	• Integrated with database and other planning, scheduling and tracking systems:financial services (banking, e-commerce)
• High performance not a major requirement	• High performance and continuous availability is a necessity • Deployed in mission-critical applications

Ref: Y. Deshpande Et Al.,"Web Engineering", J. Web Eng. 1(1): 3-17 (2002)

The World Wide Web: intro

III: Web applications categories and complexity

Characteristics of **simple** vs **advanced** Web applications (2/2)

Simple Web-based systems	Advanced Web-based systems
• Developed by single individuals or very small teams	• May require a larger development team with expertise in diverse areas
• Security requirements minimal	• Calls for risk or security assessment and management
• Easy to create and maintain	• Necessitate project plan and management • Requires a sound development process and methodology
• Feedback from users either unnecessary or not sought	• User satisfaction vital
•Web site mainly as an “identity” for the current clientele, rather as communication medium	• Web site/application as the main communication medium between the organization and users

Ref: Y. Deshpande Et Al.,”Web Engineering”, J. Web Eng. 1(1): 3-17 (2002)

The World Wide Web: intro

III: Web applications categories and complexity

Categories of web applications

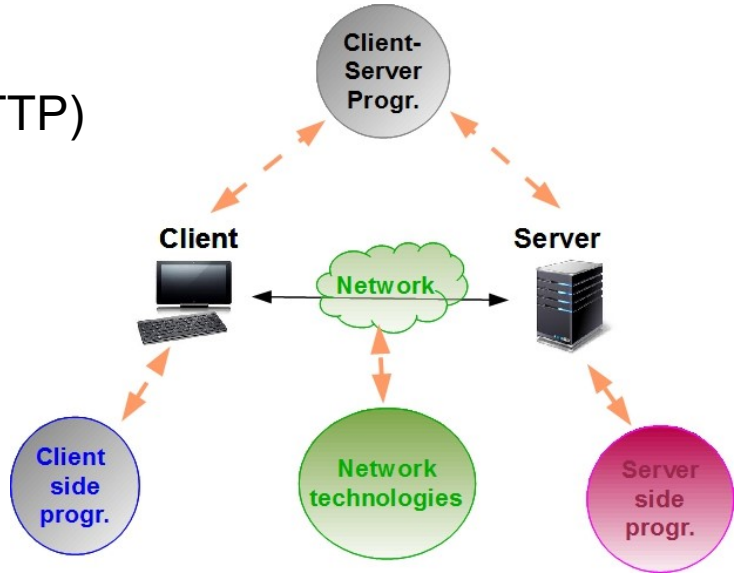
Category	Examples
• Informational	• Online newspapers, product catalogues, newsletters, classifieds, e-books
• Interactive User-provided information Customized access	• Registration forms, customized information presentation, games
• Transaction	• E-shopping, ordering goods and services, banking
• Collaborative work environments	• Distributed authoring systems, collaborative design tools
• Online communities, marketplaces	• Chat groups, recommender systems, marketplaces, social networks
• Web portals and services	• Electronic shopping malls, intermediaries

Ref: Y. Deshpande Et Al.,"Web Engineering", J. Web Eng. 1(1): 3-17 (2002)

The World Wide Web: intro

IV: Disciplines and technologies involved in the web

- Networking technologies
  - Internet layers and protocols (IP,TCP,UDP,HTTP)
  - Addressing/routing
  - Multimedia standards (A/V streaming)
  - Communication over IP (VoIP, Video conf.)

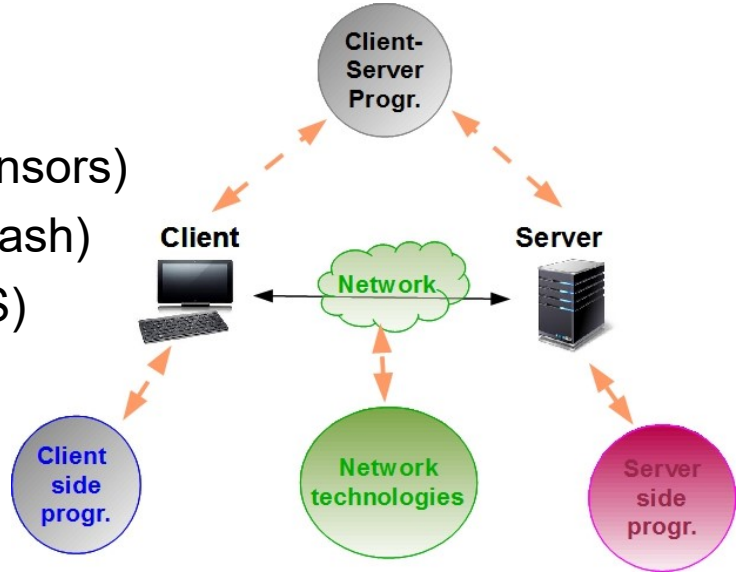


- Client-server programming
  - Hypertext systems
  - Distributed systems
  - HTTP protocol
  - Programming paradigms (Ajax, Mashup, Model-view-controller)
  - Software engineering

The World Wide Web: intro

IV: Disciplines and technologies involved in the web

- Client-side programming
  - HCI (GUI design, browsers, interaction design)
  - Multimodal interaction (audio, video, speech, sensors)
  - Scripting languages (JavaScript, Java Applet, Flash)
  - Hypermedia languages (HTML, XML, DTD, CSS)

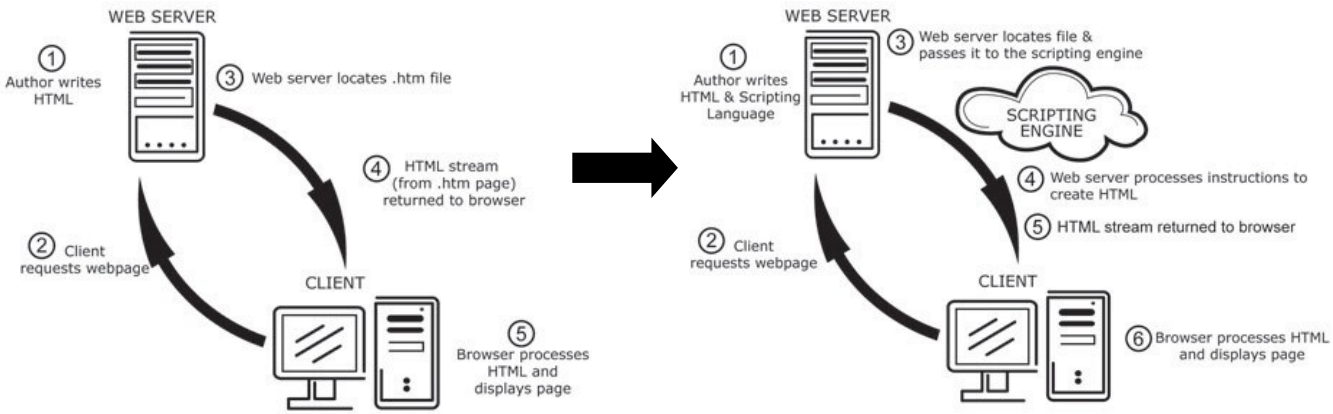


- Server-side programming
  - Multithread programming
  - Database systems and lang. (MySql, Query Lang, XML)
  - Dynamic web applications (Java Servlet, JSP, PHP)
  - Multimedia codecs and transmission standards (Jpeg/Mpeg)
  - Distributed computing, cloud computing, etc.

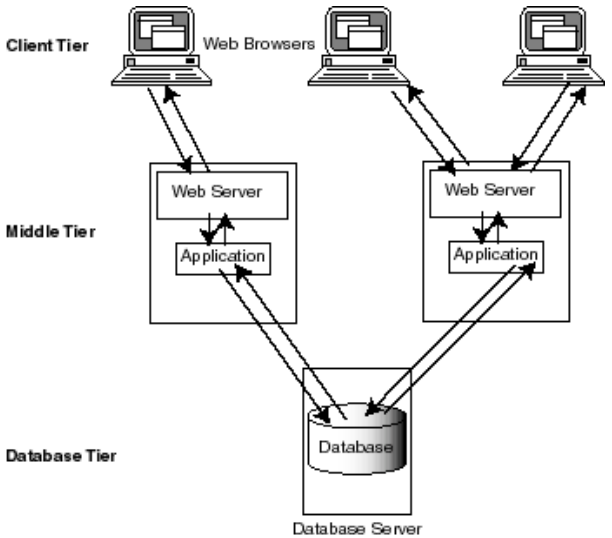
The World Wide Web: intro

VI: Some of today's web design patterns and trends

- Dynamic web pages (vs static web pages)



- Database oriented web applications

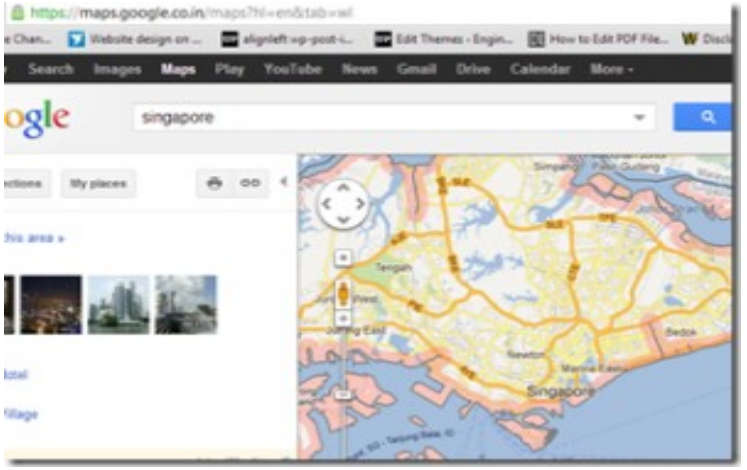


Credits: pictures by http://blog.search3w.com/

The World Wide Web: intro

VI: Some of today's web design patterns and trends

- Local page component update (vs global page update): Ajax programming



- Responsive web design

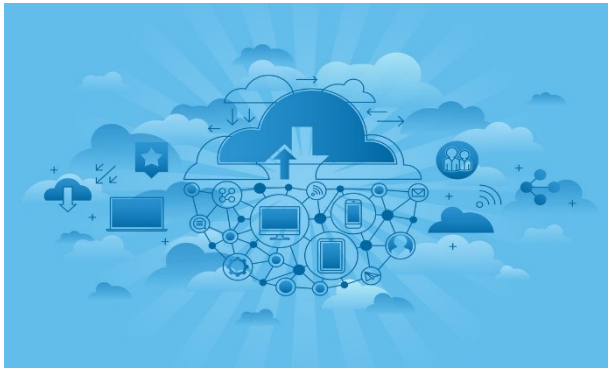


Credits: pictures by http://www.thesearchagents.com/

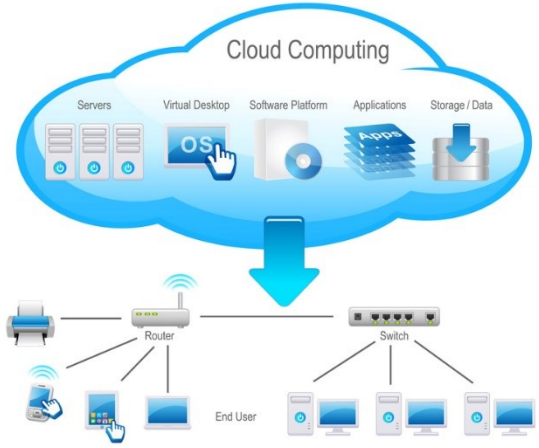
The World Wide Web: intro

VI: Some of today's web design patterns and trends

- Cloud data hosting e
  Web services



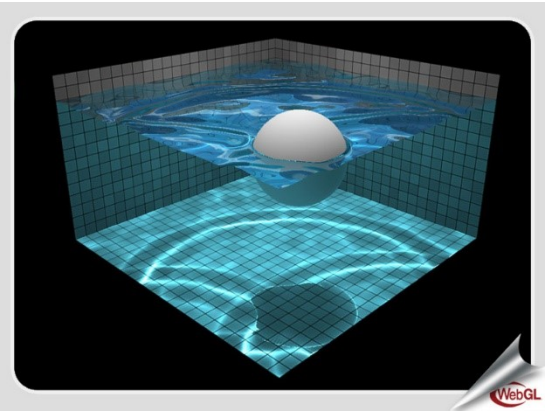
- Cloud computing



The World Wide Web: intro

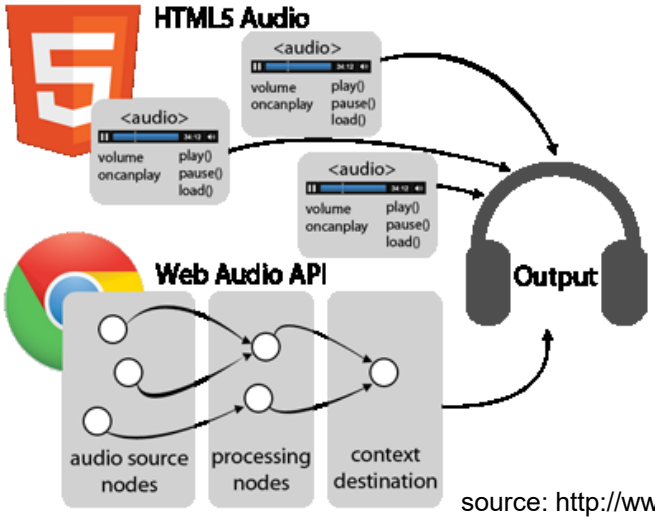
V: Latest interactive multimedia technologies for the web

- Interactive graphics
  - WebGL: javascript library providing OpenGL functionalities



WebGL Water demo by Evan Wallace: http://madebyevan.com/webgl-water/

- Interactive audio
  - WebAudioApi: javascript libra for audio manipulation

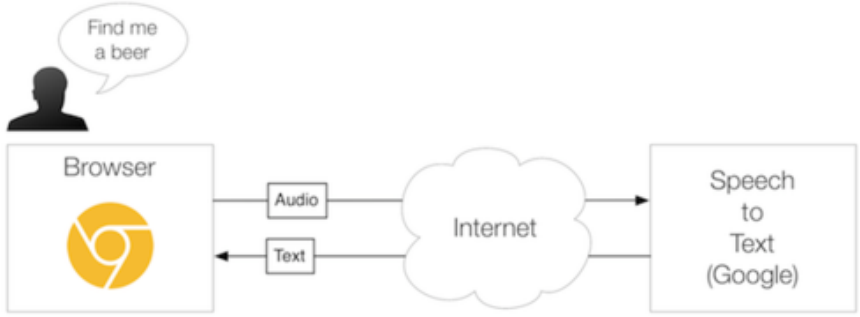


source: http://www.devbattles.com/

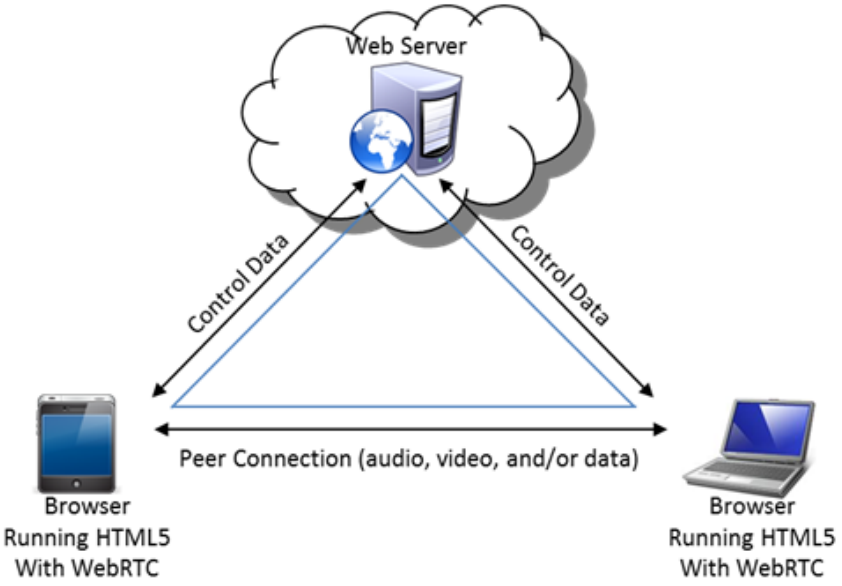
The World Wide Web: intro

V: Latest interactive multimedia technologies for the web

- Speech based interaction
  - WebSpeechAPI: library for ***speech recognition*** and ***speech synthesis*** (TTS)



- Audiovisual streaming for peer-to-peer communication
  - WebRTC: point-to-point and multipoin Real Time Communication protocol (e.g., videoconference)

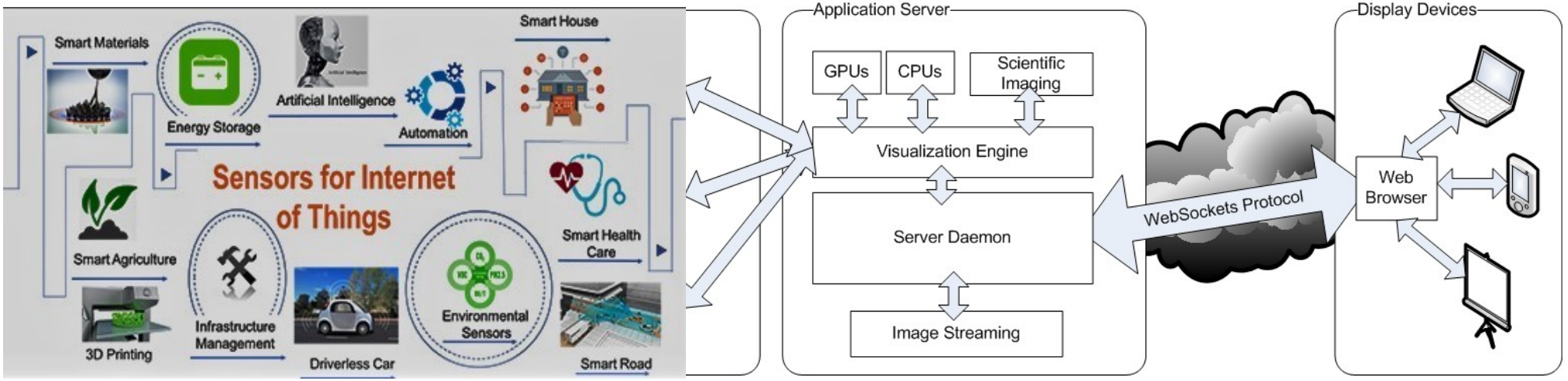


source: http://www.nojitter.com/

The World Wide Web: intro

V: Latest interactive multimedia technologies for the web

- Real time data visualization for distributed sensor networks

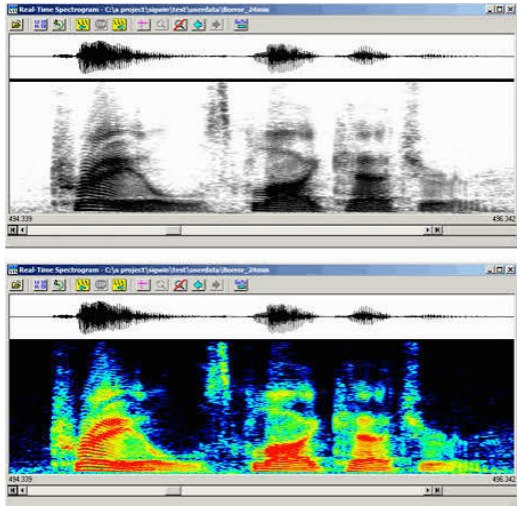


The World Wide Web: intro

V: Latest interactive multimedia technologies for the web

- Dashboard design for real time data visualization

Audio/speech data



Stock market data



Sensor data

